

5. (3 points)

Given the following information as of December 31, 2016:

Accident <u>Year</u>	Onlevel <u>Premium</u>	Cumulative <u>Paid Loss</u>
2014	\$400,000	\$210,000
2015	375,000	130,000
2016	450,000	50,000

- $G(x) = \frac{x^{1.5}}{x^{1.5} + 15^{1.5}}$, where G is the cumulative proportion of ultimate losses paid and x the average age in months.
- Parameter standard deviation for Cape Cod method: 175,000
- Process variance/mean scale parameter (σ^2) for Cape Cod method: 3,000

a. (1.5 points)

Calculate the total standard deviation of the Cape Cod method's total loss reserve indication.

b. (1 point)

Calculate the total loss reserve by credibility-weighting the two indications from the Cape Cod method and chain ladder method using the Benktander method.

c. (0.5 point)

Identify and briefly describe a different growth curve form that would be more appropriate to approximate the loss payment pattern for a short-tailed line of business.

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EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 5				
TOTAL POINT VALUE: 3			LEARNING OBJECTIVE(S): A2	
SAMPLE ANSWERS				
Part a: 1.5 points				
<u>Sample 1</u>				
AY	Avg Age	G(X)	CC Reserve = (1-G(x))*ELR*Prem	
2014	30	.739	67,986	
2015	18	.568	105,462	
2016	6	.202	233,694	
Total			407,106	
$ELR = \frac{210,000 + 130,000 + 50,000}{400,000(.739) + 375,000(.568) + 450,000 (.202)} = 0.651$				
Process Variance = 3,000 * 407,106				
Parameter Variance = 175,000 ^2				
Total Std Dev = $\sqrt{(Proc Var + Param Var)} = 178,456$				
<u>Sample 2</u>				
AY	Avg Age	G(X)	CC Reserve = (G(66)-G(x))*ELR*Prem	
2014	30	.7388	42,529	
2015	18	.5679	81,573	
2016	6	.2019	205,058	
Total			329,161	
Truncation = G(66) = 0.9022				
$ELR = \frac{210,000 + 130,000 + 50,000}{400,000(.7388) + 375,000(.5679) + 450,000 (.2019)} = 0.6507$				
Total Std Dev = $\sqrt{(3,000 * 329,161 + 175,000 ^2)} = 177,799$				
Part b: 1 point				
<u>Sample 1</u>				
AY	CL Res = Pd/G(X) - Pd	Z = G(X)	Cred Res = Z * CL Res + (1-G(X))* CC Res	
2014	74,246	.739	72,611	
2015	98,894	.568	101,716	
2016	197,642	.202	226,415	
			400,742	
<u>Sample 2</u>				
CL Res = Paid * (G(66)/(G(x) - 1)				
AY	CL Res	Z = G(66)/G(X)	Cred Res = Z * CL Res + (1-G(X))* CC Res	

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2014	46,445	.819	45,736
2015	76,526	.629	78,398
2016	173,427	.224	197,973
			322,108

Part c: 0.5 point

Sample 1

Weibull growth curve $G(x) = 1 - e^{-\left(\frac{x}{\theta}\right)^\omega}$ has a lighter tail, so development is completed more quickly which is more appropriate for a short tail LOB

Sample 2

For a short-tailed line of business, the growth curve can be truncated so that it will not further develop after a certain period.

EXAMINER'S REPORT

Candidates were expected to demonstrate knowledge of the Cape Cod reserve method, including the calculation of the standard deviation of the reserves and relationship of credibility to the Benktander method.

Part a

Candidates were expected to calculate the Cape Cod reserve and the standard deviation of those loss reserves.

A common error was calculating the ELR reserve by year instead of the Cape Cod reserve.

Part b

Candidates were expected to calculate the chain ladder reserves and credibility weight those reserves by accident year with the Cape Code reserves.

Common errors include:

- Using an incorrect credibility formula (squaring the credibility factor or multiplying by the ELR)
- Swapping the credibility applied to each reserve
- Using something other than the chain ladder or Cape Cod loss reserves as stated.

Part c

Candidates were expected to identify a different growth and describe why it is more appropriate for a short tailed line of business.

Common errors include:

- Listing a different curve without explanation.
- Selecting different parameters of the log-logistic growth curve provided that were not determined by maximum likelihood.
- Incorrectly identifying the Weibull growth function as exponential.